

10(a)	causing X / -70 (mV) any <i>three</i> from: 1 active transport / sodium-potassium pump ; 2 moves three Na⁺ out of and two K⁺ in (to neurone) ; 3 more K⁺ <u>diffuses</u> out than Na⁺ <u>diffuses</u> in ; 4 makes outside of membrane (relatively) more positive ; 5 maintaining / establishing, resting potential ; causing Y / +40 (mV) any <i>two</i> from: 6 depolarisation ; 7 Na⁺ / sodium ion, channels open ; 8 Na⁺, moves / diffuses, in (to neurone) ; causing Z / -90 (mV) any <i>two</i> from: 9 hyperpolarisation ; 10 (voltage-gated) K⁺ / potassium ion, channels open ; 11 K⁺, moves / diffuses, out (of neurone) ;	7
10(b)	1 local, circuit / current ; 2 from, depolarised / positive, area to, resting / negative, area ;	2
10(c)	any <i>three</i> from: 1 speeds up (nerve) impulse / impulse transmission faster ; 2 (as) myelin sheath / Schwann cells, act(s) as an insulator ; 3 (as) local circuits flow between, nodes of Ranvier / gaps in sheath ; 4 saltatory conduction ; 5 fast(er), (described) response / muscle contraction ;	3